DO NOW — RECALL

# Testing and program quality

Name: _____ Class: _____ Date: _____

**1. Why should Largest() require a nonempty input array?**

**2. What is a regression?**

**3. What does a class method do?**

**4. A date validator passes all unit tests but booking fails when writing to the database. What test is needed?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It reads a[0] as its starting value.

### 2. Model answer

Previously working functionality fails after a change.

### 3. Model answer

Defines behaviour for objects of that class.

### 4. Model answer

An integration test of the database interaction.

### Vocabulary — know these terms

unit testing — Test one module/function in isolation.

integration testing — Check interaction of components.

system testing — Test the entire application as a system.

acceptance testing — Check fitness for user/business requirements.

regression testing — Repeat tests to catch defects after changes.

NEW LEARNING

# Read, annotate and explain

### Test individual units

Unit tests isolate one function or class method from the rest of the system.

### Test connections and behaviour

Integration tests interactions between modules; system tests the complete application.

### Confirm intended outcome

Acceptance testing checks whether the delivered application meets stakeholder requirements. Automated regression tests rerun after changes.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### A checkout computes the right total but fails to save orders. Which test is most relevant?

### Predict first

TotalCalculator already gives correct values; the failure appears when OrderService interacts with the database.

### Model and explain

Prioritise an integration test of saving, then a system-level end-to-end checkout test.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Match each: individual ValidDate function, database interaction, full website journey, client sign-off to the appropriate test level.

### Question 2

A login test passes until a new password rule is introduced, after which registration fails. Explain a sensible debugging and testing workflow.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Why is 100% execution of code lines not equivalent to proving an application correct? Suggest three further forms of evidence.

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Passing unit tests alone proves that the complete application works for its users.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.